

Maestro Cloud Control

Enabling Access to Azure

Subscriptions

Admin Guide

M3AG-1

October 2024

Version 3.3

CONTENTS

Preface	3
About This Guide.....	3
Audience.....	3
Pre-Requisites	3
1 Background Information	4
2 Preparing for Connection	5
2.1 Creating a Microsoft Entra ID Application and Service Principle.....	5
2.2 Billing Export Setup	6
2.2.1 Configuring Storage Account for Billing Export	6
2.2.2 Creating a Role for Billing Application	11
2.3 Grant Permissions for the Service Principal	14
2.4 Completing SSO Configuration	15
2.4.1 Receiving URL Endpoints for SSO	15
2.4.2 Setting Approved URL	16
2.5 Add API Permissions	16
2.6 Generate a Client Secret for the Microsoft Entra ID Application:	19
3 Share the connection details with Maestro Team	20
Version History	21

PREFACE

ABOUT THIS GUIDE

This instruction provides the information on the pre-requisites and steps that should be done to connect a new customer Azure subscription to Maestro.

AUDIENCE

The guide is designed for administrators of Azure tenants and subscriptions, who want to enable login using their Microsoft Entra ID (Azure AD) and enable billing and management access for the Maestro application.

PRE-REQUISITES

Before starting the activation process, please insure you have the following:

- At least `Application Developer` access level to the Microsoft Entra ID Tenant in which the application will be created.
- An `Owner` access level to the Azure Subscription that you want to connect to Maestro.
- Access to the Azure portal or a local installation of the Azure CLI.

Also make sure that the following providers are activated in each subscription which Maestro will be working with:

- Microsoft.Compute
- Microsoft.Network
- Microsoft.Security
- Microsoft.Storage
- Microsoft.CostManagementExports

For example, to activate the Microsoft.Network provider, use the following Azure CLI command:

```
az provider register --namespace Microsoft.Network
```

1 BACKGROUND INFORMATION

Maestro allows users to utilize single sign-on (SSO) login via Microsoft Entra ID (Azure AD), manage virtual infrastructures, and receive billing information for connected clouds.

To enable the necessary functionality on Azure subscriptions, the respective configurations should be completed within the Azure tenant and subscriptions that Maestro will interact with.

Enabling SSO, Management, and Billing functionalities is accomplished by properly configuring Microsoft Entra ID applications. This can be achieved either through a single application that consolidates all necessary permissions or through multiple applications.

This instruction provides detailed step-by-step guidance and a list of items to be shared with the Maestro setup team to complete the connection.

2 PREPARING FOR CONNECTION

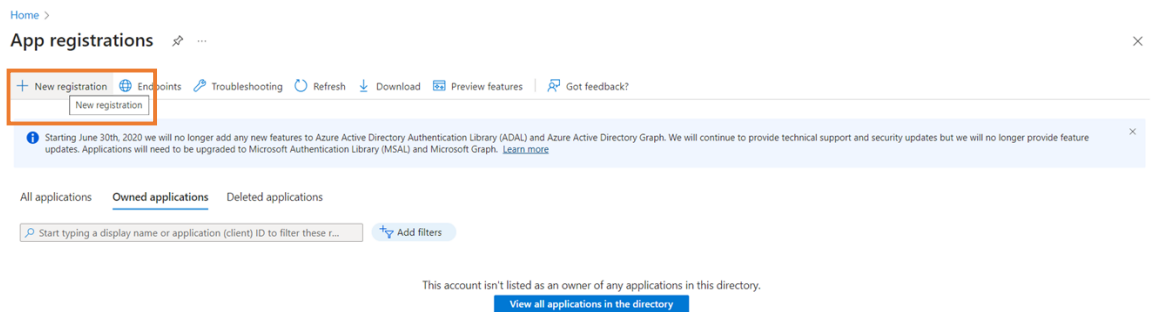
To be able to use SSO login to Maestro via Microsoft Entra ID or to connect an Azure subscription, you need to set up a Microsoft Entra ID application, a service principal, and generate a Client Secret.

If you want to use different applications for Login access, Management, and Billing, you will need to repeat application creation steps respectively.

2.1 CREATING A MICROSOFT ENTRA ID APPLICATION AND SERVICE PRINCIPLE

To create the Application in the Azure Portal:

1. Go to the **Microsoft Entra ID** overview on the **Azure Portal** and select the **App Registration** blade.
2. Click **“New registration”** to add a new Microsoft Entra ID Application:



3. In the form that opens, set the following values and press **“Register”**:

- **Name** - The user-facing display name for this application (e.g., `maestro-service`).
- **Supported Account Types** - it is recommended to set to "Accounts in this organizational directory only (single-tenant)".
- **Redirect URI** - select "Web" for the URI type and leave the value blank:

Home > App registrations >

Register an application

* Name

The user-facing display name for this application (this can be changed later).

maestro-service ✓

Supported account types

Who can use this application or access this API?

☒ Accounts in this organizational directory only (Single tenant)

☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)

☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant) and personal Microsoft accounts (e.g. Skype, Xbox)

☐ Personal Microsoft accounts only

[Help me choose...](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Web e.g. https://example.com/auth ✓

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

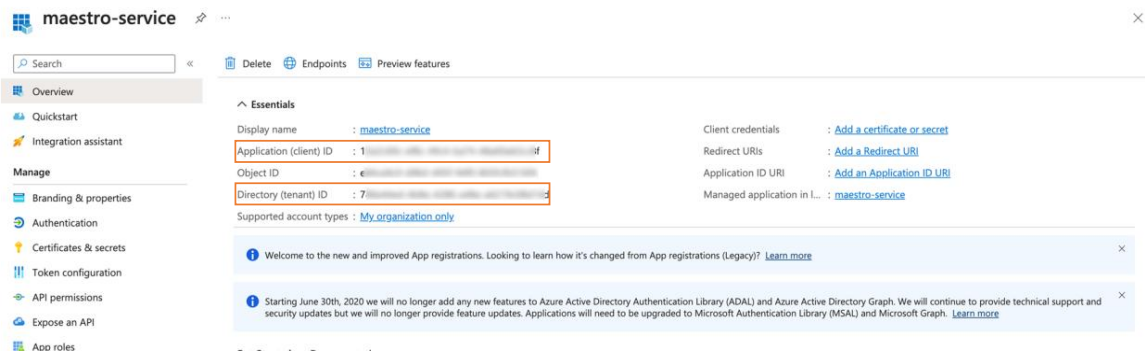
By proceeding, you agree to the [Microsoft Platform Policies](#)

Register

After pressing “**Register**”, a new Microsoft Entra ID application is created and displayed to the screen. If this does not happen, navigate to the “**App Registration**” blade, select your Microsoft Entra ID application from the list and click on its name.

At the top of the page with your registered application, in the “**Essentials**” section, find and copy for further usage the following application properties:

- Application (client) ID
- Directory (tenant) ID



Alternatively to this flow, you can use Azure CLI to [create Microsoft Entra ID application](#) and [Service Principal](#):

2.2 BILLING EXPORT SETUP

Enabling billing export from an Azure account needs the following steps:

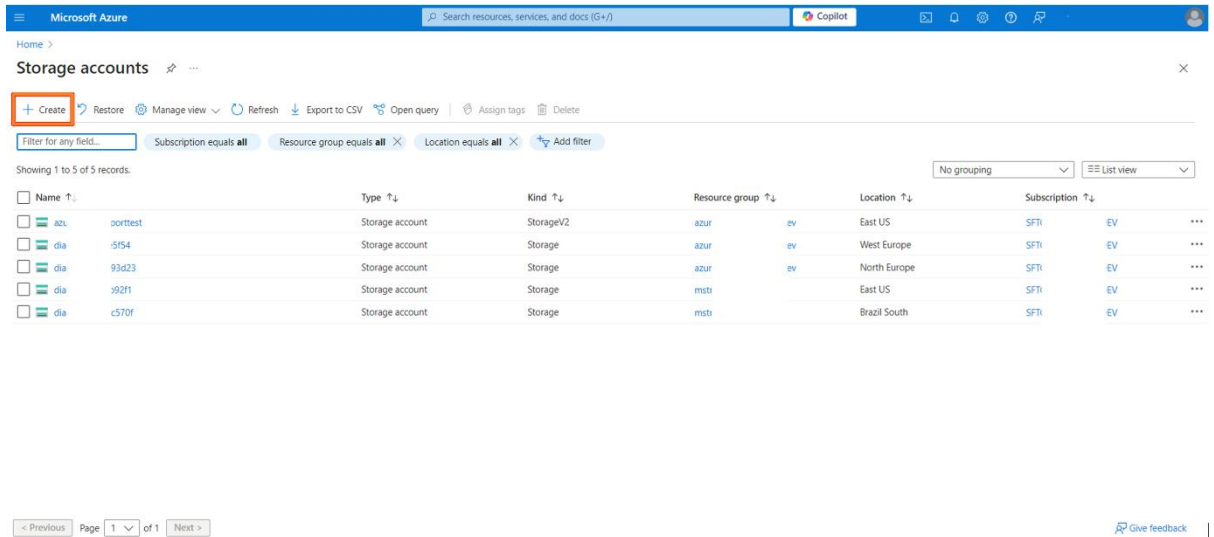
- Configuring a storage account for billing export.
- Creating a role for billing applications.
- Assigning billing export role to billing application.

2.2.1 Configuring Storage Account for Billing Export

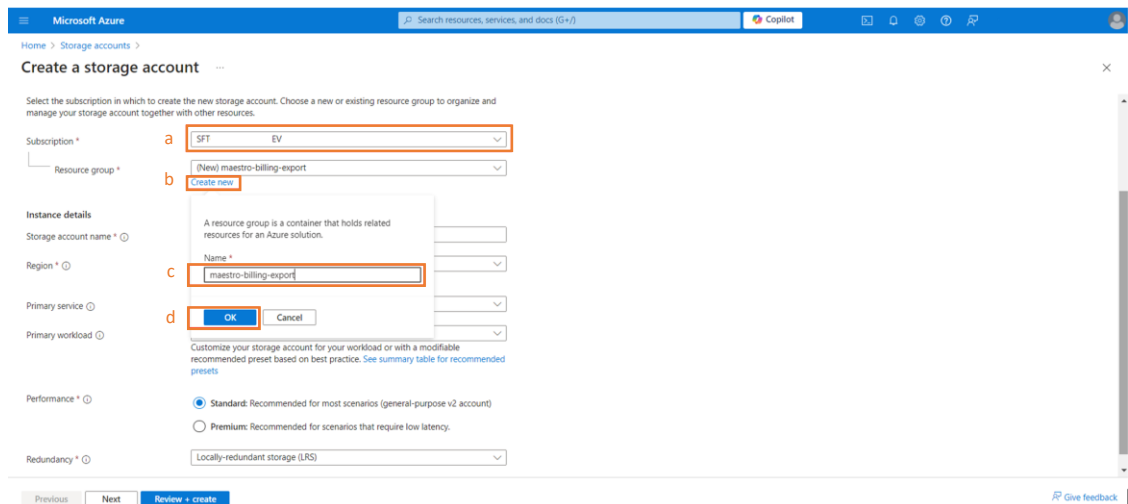
To configure the storage account for billing export, do the following:

1. In the **Storage Accounts** on the **Azure Portal**, click **Create**:

Maestro – Enabling Access to Azure Subscriptions



2. Select the required **Subscription (a)**, then click **Create new (b)**. In the opened wizard, type a resource group **Name**, for example **maestro-billing-export (c)**, then click **OK (d)** to create the resource group and close the wizard:



3. Create a unique **Storage account name (e)** and select the required **Region (f)**. Set **Primary service** to “*Azure Blob Storage or Azure Data Lake Storage Gen 2*” (**g**), then Click **Review + create (h)**.

Maestro – Enabling Access to Azure Subscriptions

Microsoft Azure

Home > Storage accounts >

Create a storage account

manage your storage account together with other resources.

Subscription * SFT EV

Resource group * (New) maestro-billing-export
[Create new](#)

Instance details

Storage account name * e maestrobillingexport

Region * f (Asia Pacific) Central India
[Deploy to an Azure Extended Zone](#)

Primary service g Azure Blob Storage or Azure Data Lake Storage Gen 2

Primary workload Select...

Performance * Standard: Recommended for most scenarios (general-purpose v2 account)
Premium: Recommended for scenarios that require low latency.

Redundancy * Locally-redundant storage (LRS)

Previous Next **Review + create** h

[Give feedback](#)

- Once you checked the correctness of all the entered parameters, click **Create** to create a new deployment, which will create your storage account:

Microsoft Azure

Home > Storage accounts >

Create a storage account

Basics Advanced Networking Data protection Encryption Tags **Review + create**

[View automation template](#)

Basics

Subscription SFT EV

Resource group maestro-billing-export

Location Central India

Storage account name maestrobillingexport

Primary service Azure Blob Storage or Azure Data Lake Storage Gen 2

Primary workload

Performance Standard

Replication Locally-redundant storage (LRS)

Advanced

Enable hierarchical namespace Disabled

Enable SFTP Disabled

Enable network file system v3 Disabled

Allow cross-tenant replication Disabled

Access tier Hot

Enable large file shares Enabled

Previous Next **Create** h

[Give feedback](#)

- Once the deployment is completed, click **Go to resource** to open the created storage account.

Microsoft Azure

Home >

maestrobillingexport_172i 19 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name: maestrobillingexport_172i 19 Start time: 13/09/2024, 14:46:27 Correlation ID: cb5f 1c70

Subscription: SFT EV

Resource group: maestro-billing-export

Deployment details

Next steps

Go to resource

Give feedback
[Tell us about your experience with deployment](#)

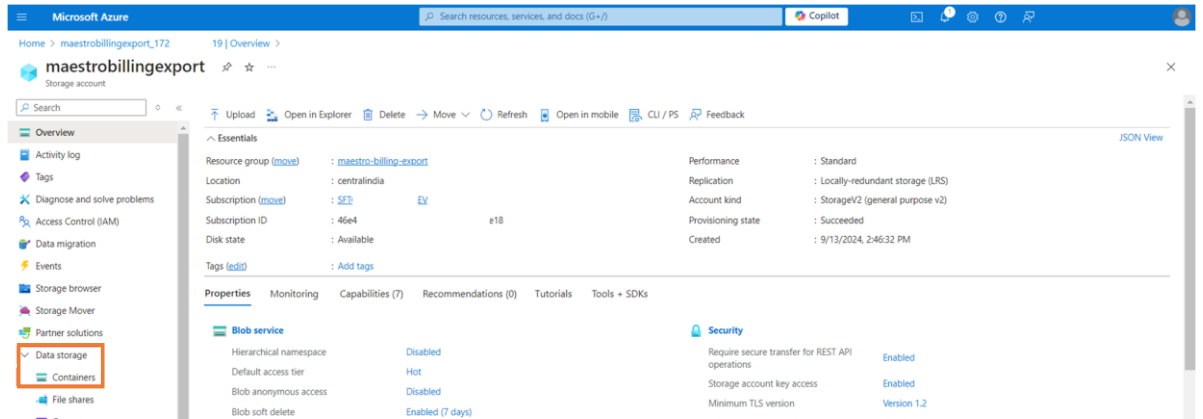
Cost Management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

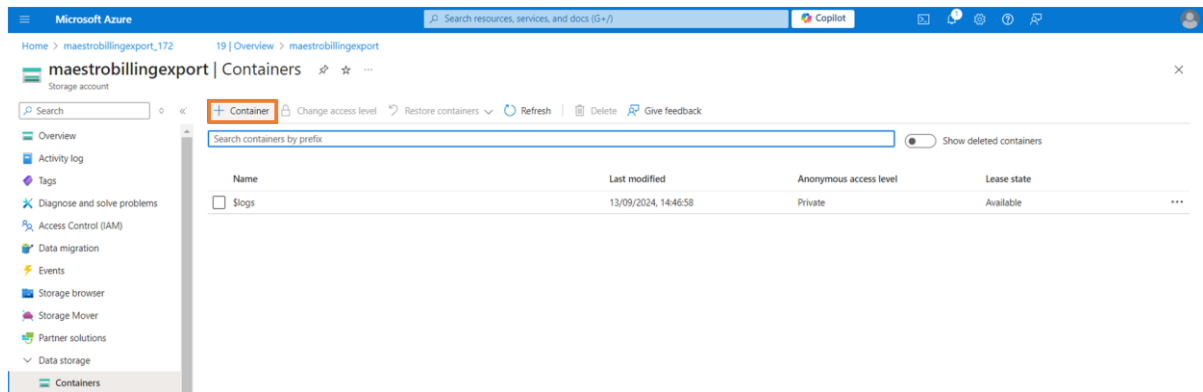
Save the created storage account name for further sharing with the Maestro team.

- In the opened page, click **Data storage->Containers**

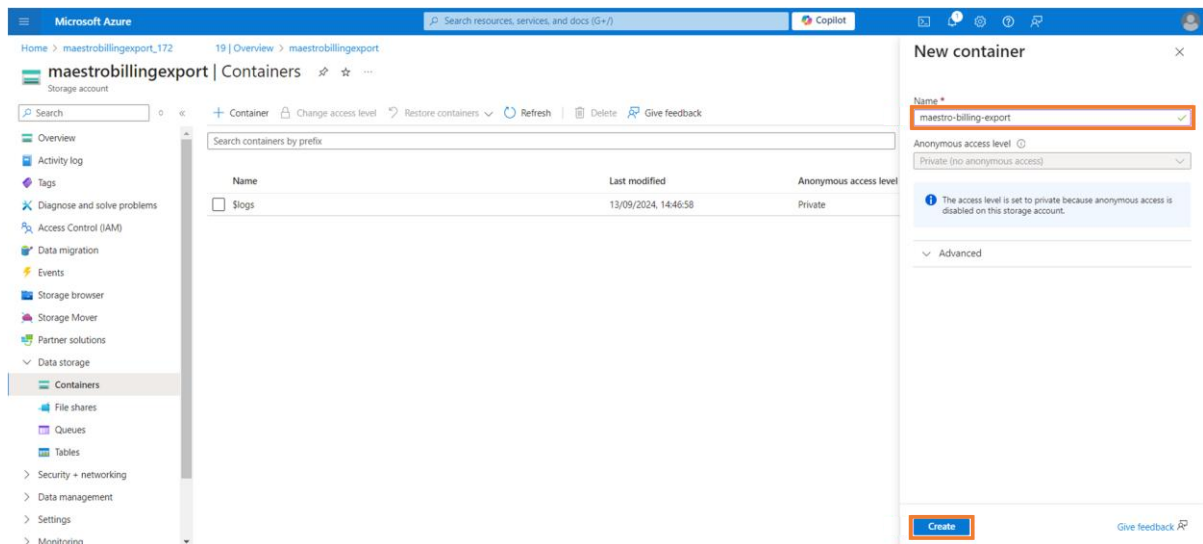
Maestro – Enabling Access to Azure Subscriptions



7. In the opened tab, click **+ Container** to open the wizard:



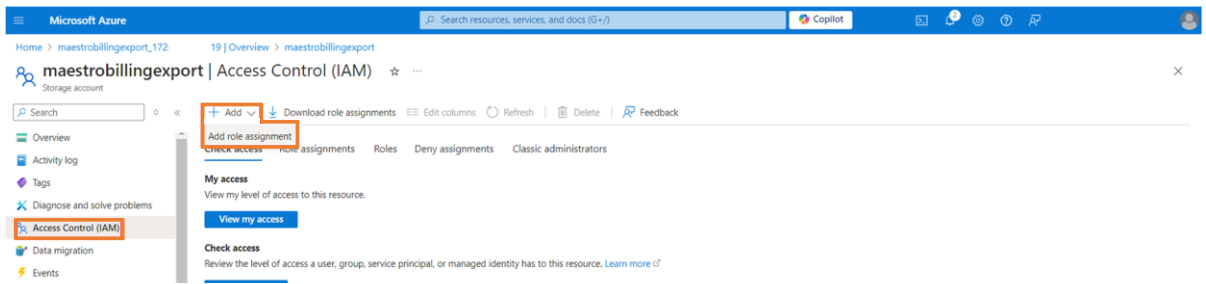
8. Set the name of the container like “*maestro-billing-export*” and click **Create**:



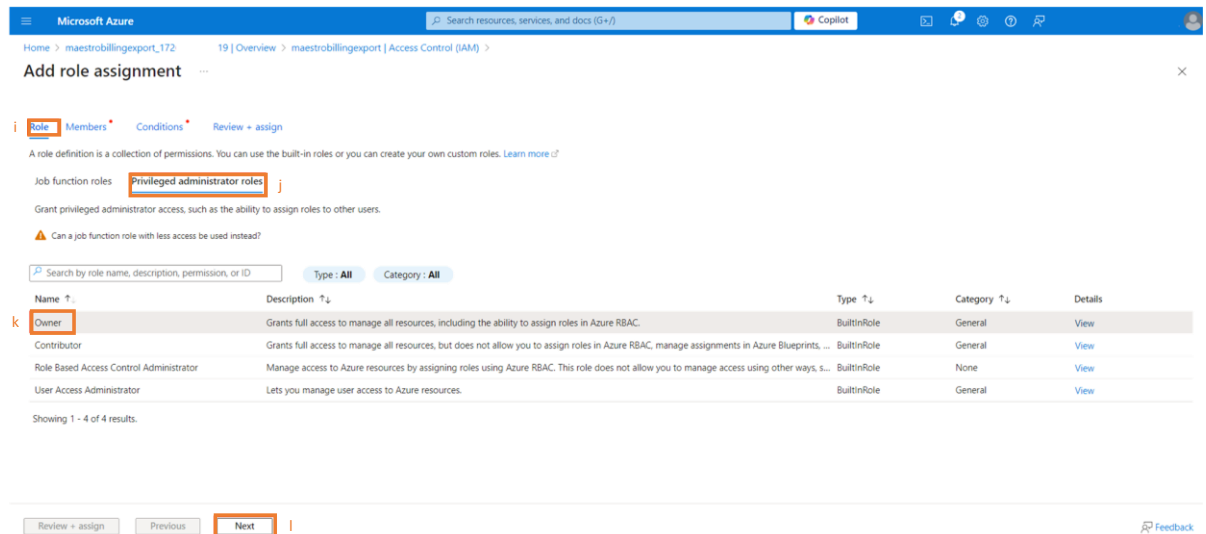
Save the created container name for further sharing with the Maestro team.

9. After the container is created, navigate to **Access Control (IAM)**, then click “**+ Add -> Add role assignment**”

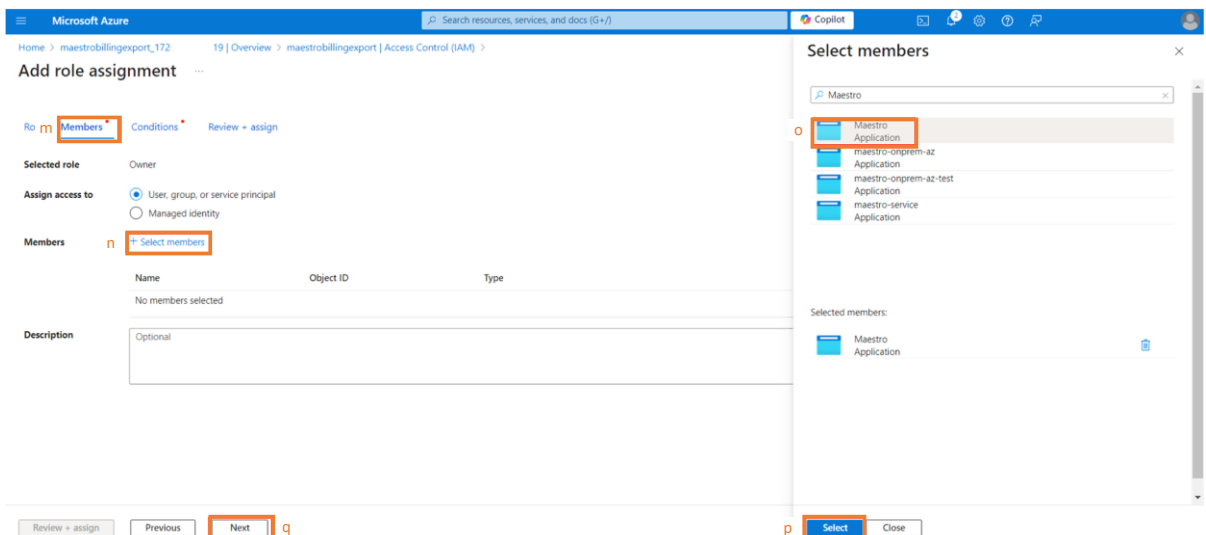
Maestro – Enabling Access to Azure Subscriptions



10. Go **Role(i) -> Privileged administrator roles (j)**, select the **Owner role (k)** in the table, and click **Next (l)**



11. Now you should be on **Members** tab (m). Click **+ Select members (n)**. In the opened wizard, click the **Maestro** application (o), and click **Select (p)** to close the wizard. Then click **Next (q)**



12. Now you should be on the **Conditions** tab (r). Select **Allow user to assign all roles (highly privileged) (s)** and click **Next (t)** :

Maestro – Enabling Access to Azure Subscriptions

Home > maestrobillingexport_1726227930419 | Overview > maestrobillingexport | Access Control (IAM) >

Add role assignment

Role Members **Conditions** Review + assign

Selected role Owner

What user can do

- ☐ Allow user to only assign selected roles to selected principals (fewer privileges)
- ☐ Allow user to assign all roles except privileged administrator roles Owner, UAA, RBAC (Recommended)
- ☒ Allow user to assign all roles (highly privileged)

Owner is a privileged admin role that grants privileged administrator access, such as the ability to assign roles to other users. Microsoft recommends that you add a condition to narrow the permissions of this role to least privilege.

Review + assign Previous **Next**

13. Carefully review all the selected parameters. If the parameters are correct, click **Review + assign** to grant the required permissions for Maestro.

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > maestrobillingexport_172 19 | Overview > maestrobillingexport | Access Control (IAM) >

Add role assignment

Role Members Conditions **Review + assign**

Role Owner

Scope /subscriptions/46e e18/resourcegroups/maestro-billing-export/providers/Microsoft.Storage/storageAccounts/maestrobillingexport

Members

Name	Object ID	Type
Maestro	c0af	50c App

Description No description

Condition None

Review + assign Previous Next

2.2.2 Creating a Role for Billing Application

To create a role for the billing application, do the following:

1. In **Subscriptions** section of the Azure Portal, find the subscription which is planned to be connected to Maestro and click its name:

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > **Subscriptions**

EPAM DEV

+ Add Manage Policies View Requests View eligible subscriptions

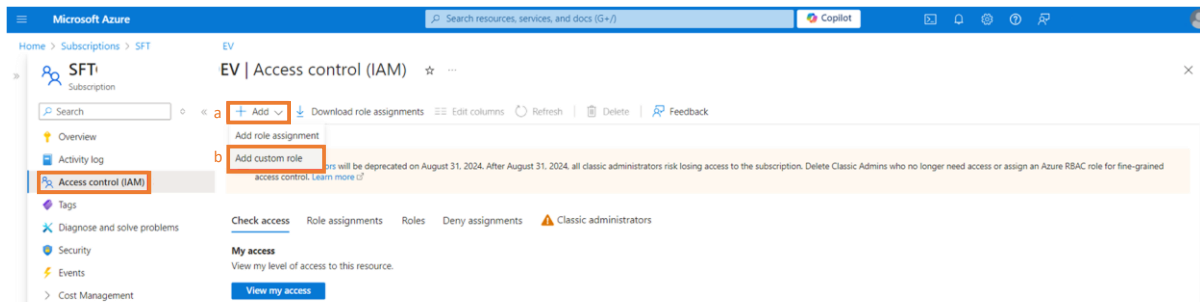
Global administrators can manage all subscriptions in this list by updating their policy setting [here](#).
Showing subscriptions in EPAM DEV directory. Don't see a subscription? [Switch directories](#)

Search for any field... Subscriptions: All (1 of 1) My role == all Status == all Add filter

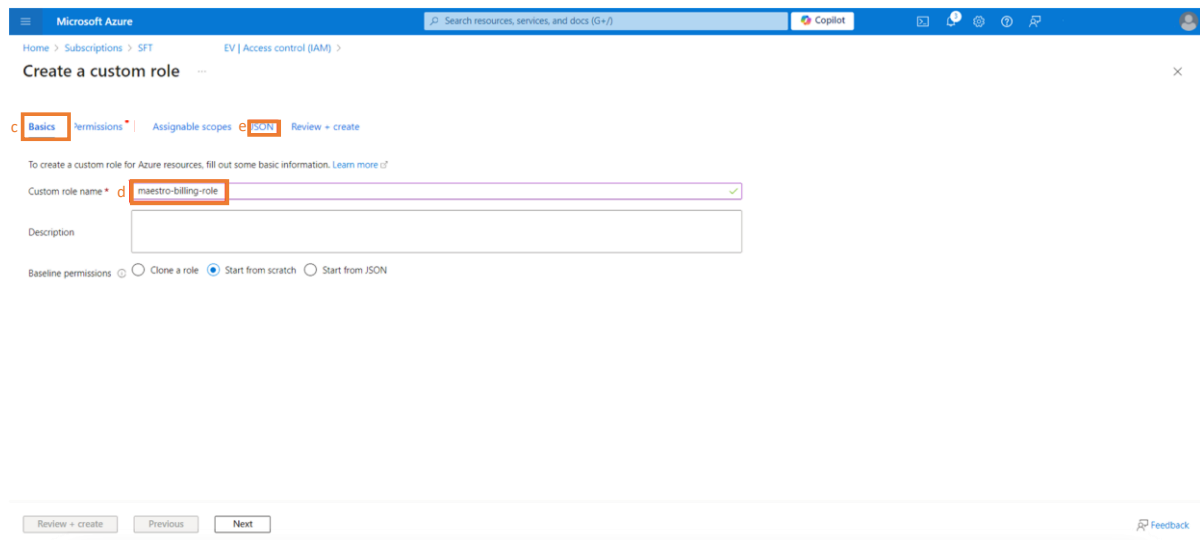
Subscription name ↑↓	Subscription ID ↑↓	My role ↑↓	Current cost	Secure Score ↑↓	Parent management group ↑↓	Status ↑↓
46e	46e	e18 Owner	-	-	Tenant Root Group	Active

Maestro – Enabling Access to Azure Subscriptions

2. Navigate to **Access control (IAM)** section. Click **+ Add** (a), then **Add custom role** (b):



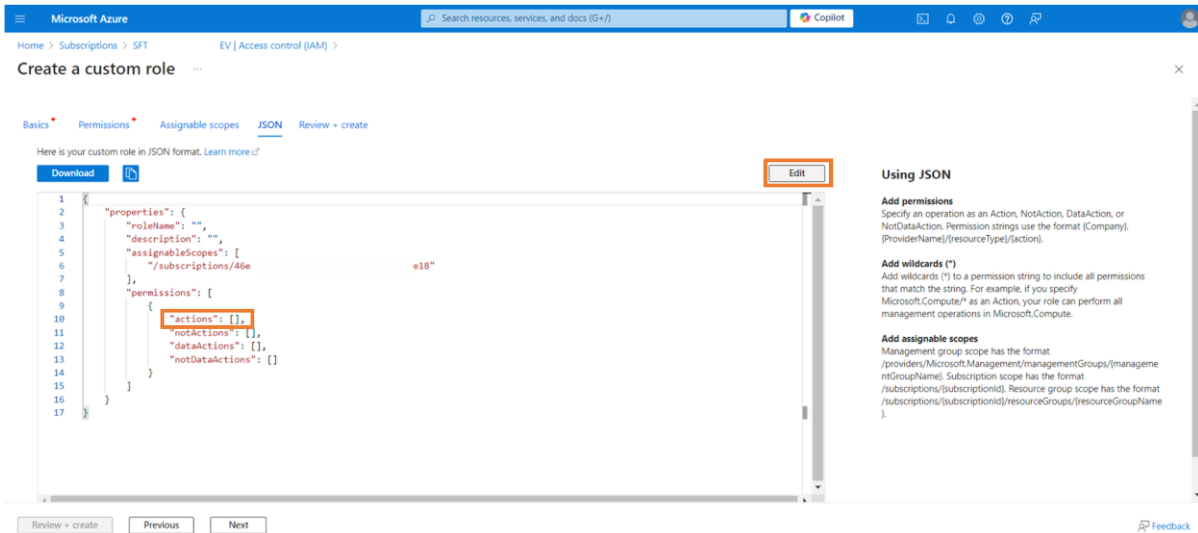
3. In the opened wizard, go to the **Basics** tab (c), set the role name (e.g. *maestro-billing-role*) (d) and navigate to the **JSON** tab (e):



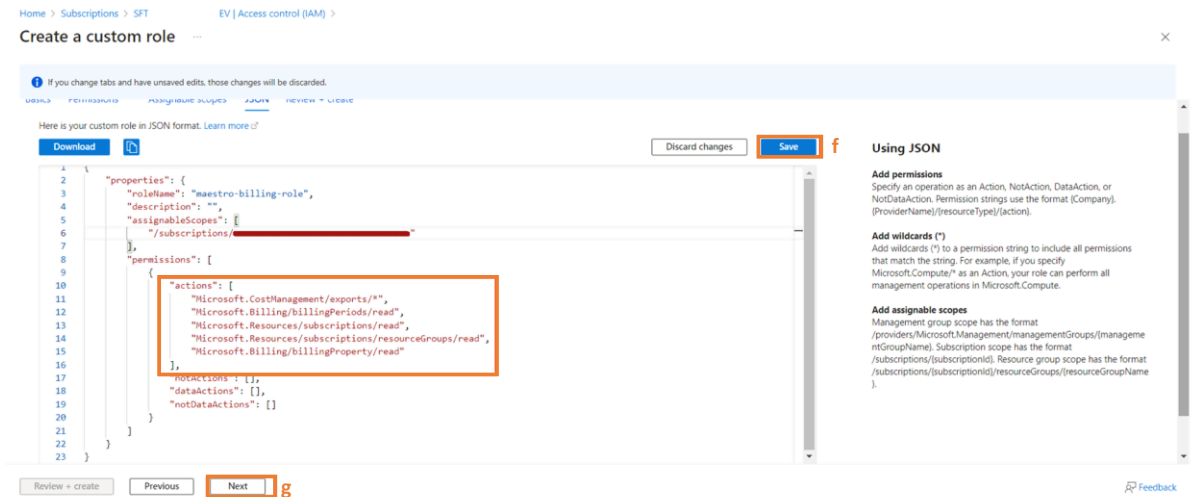
4. Click **Edit** and replace the **“actions”** element on line 10 with the following:

```
"actions": [  
    "Microsoft.CostManagement/exports/*",  
    "Microsoft.Billing/billingPeriods/read",  
    "Microsoft.Resources/subscriptions/read",  
    "Microsoft.Resources/subscriptions/resourceGroups/read",  
    "Microsoft.Billing/billingProperty/read"  
]
```

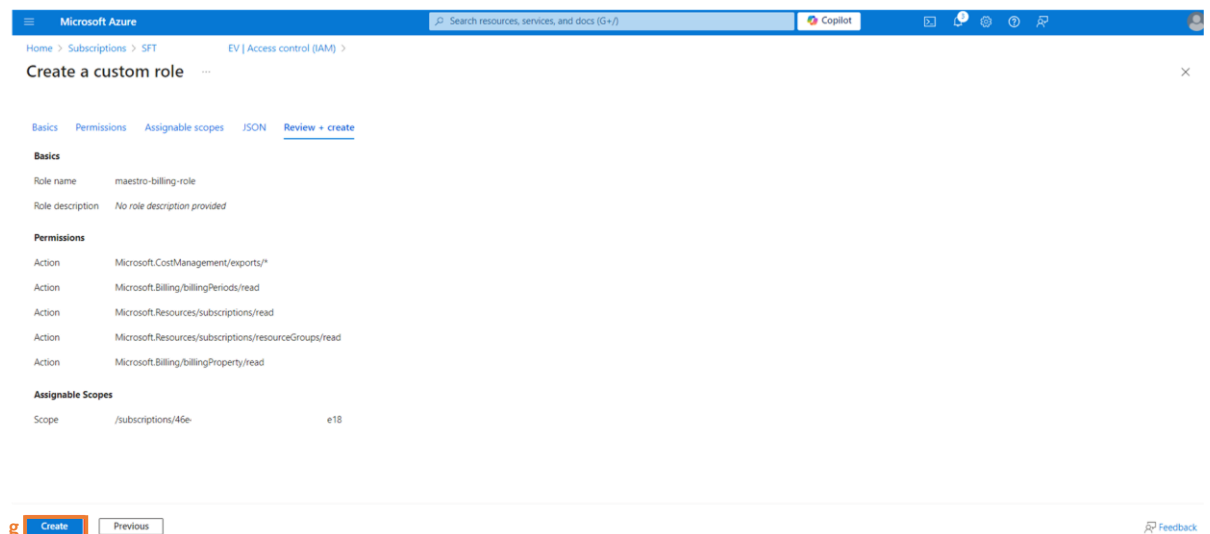
Maestro – Enabling Access to Azure Subscriptions



5. Your final role should look like on the image below. Click **Save** (f) then click **Next** (g):



6. On the **Review + create** tab, check all the entered parameters, and click **Create**:

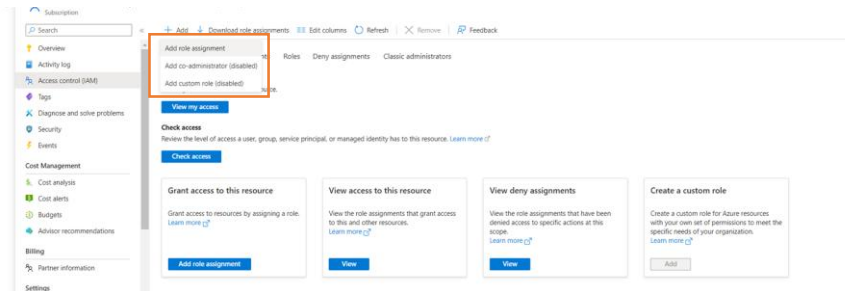


2.3 GRANT PERMISSIONS FOR THE SERVICE PRINCIPAL

To allow Application access to your Subscriptions, you need to grant necessary permissions via its linked Principle.

Please note: If you want Maestro to access several subscriptions, please repeat the steps below for each of them.

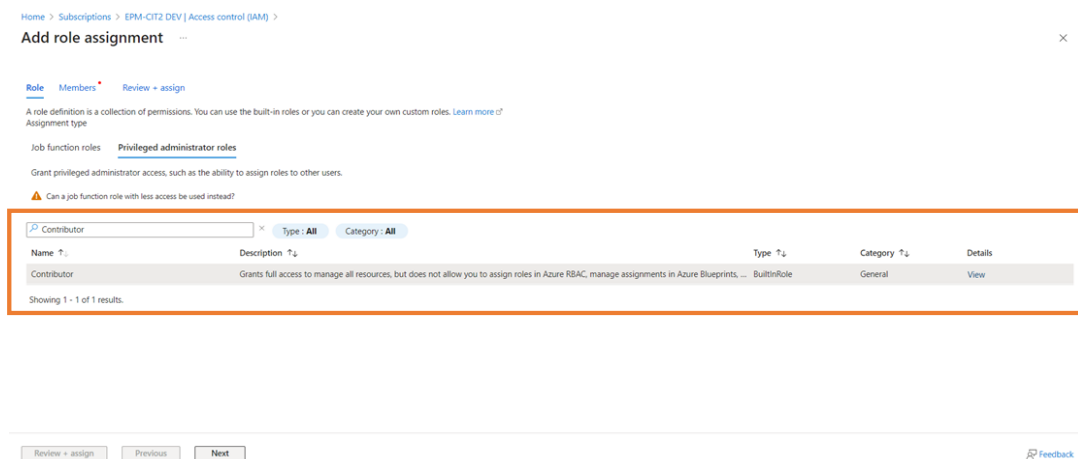
1. Go to the **Subscriptions** blade on the **Azure Portal**, select the **Subscription**, then click **Access Control (IAM) -> Add -> Add role assignment**:



2. Specify [the necessary role](#):

- **Contributor** for Management
- The role created while [Creating a Role for Billing Application](#) for Billing

You can specify both roles for the same application, or split between different ones:



3. Select the name of the created Service principal to assign this role to and click "Save":

Maestro – Enabling Access to Azure Subscriptions

Home > Subscriptions > [Subscription] > Access control (IAM) > Add role assignment

Add role assignment

Role Members **Review + assign**

Selected role Contributor

Assign access to ☒ User, group, or service principal ☐ Managed identity

Members + Select members

Name	Object ID	Type
No members selected		

Description Optional

Select members

Select

No users, groups, or service principals found.

Selected members:

- maestro-service [Remove](#)

[Review + assign](#) [Previous](#) [Next](#) [Select](#) [Close](#)



You also can use Azure CLI to create a [role assignment](#).

Now, the Service Principal should have the required permissions to the Azure Subscription

2.4 COMPLETING SSO CONFIGURATION

Configuring SSO needs two additional steps:

- Receiving URL endpoints
- Setting approved URL.

You can find the detailed instructions below.

2.4.1 Receiving URL Endpoints for SSO

The steps below are necessary for Maestro configuration team to be able to set up SSO from Maestro side.

To find the endpoints for an application you've registered, in the [Microsoft Entra admin center](#) navigate to: **Identity > Applications > App registrations > <YOUR-APPLICATION> Endpoints**:

maestro-service

Search

Overview Endpoints Preview features

Endpoints

Essentials

Display name : [maestro-service](#)

Application (client) ID : 1

Object ID : 6

Directory (tenant) ID : 7

Supported account types : [My organization only](#)

Client credentials : [Add a certificate or secret](#)

Redirect URIs : [Add a Redirect URI](#)

Application ID URI : [Add an Application ID URI](#)

Managed application in L. : [maestro-service](#)

Welcome to the new and improved App registrations. Looking to learn how it's changed from App registrations (Legacy)? [Learn more](#)

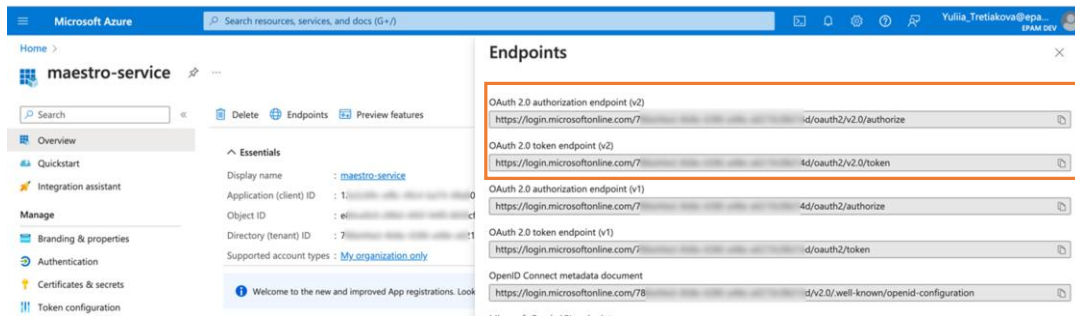
Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. We will continue to provide technical support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

From the list of the endpoints, copy for further sharing with Maestro Configuration team:

- Auth 2.0 authorization endpoint (v2)

Maestro – Enabling Access to Azure Subscriptions

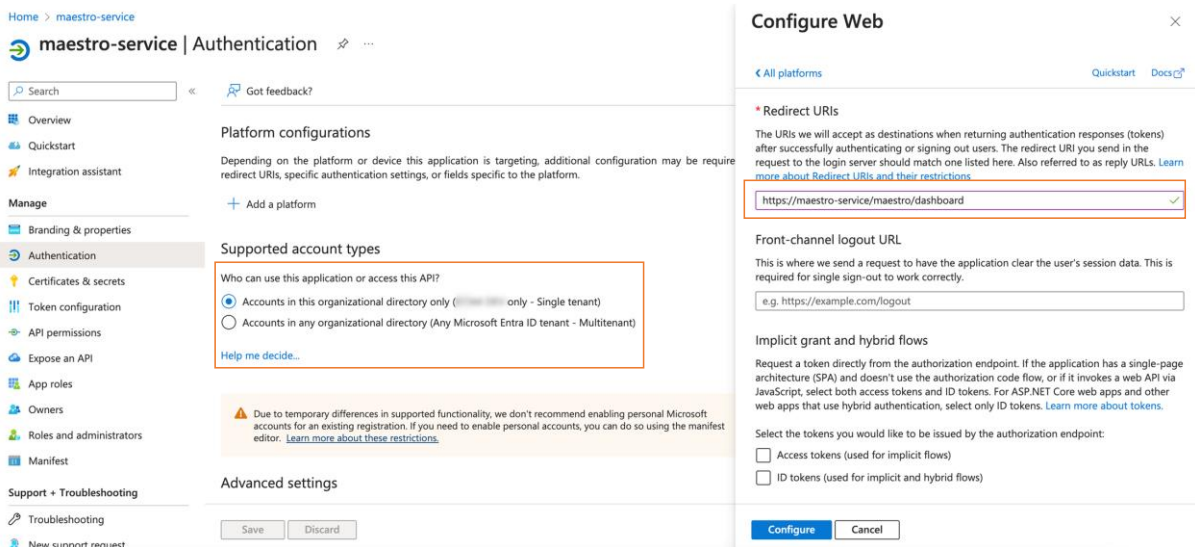
- Auth 2.0 token endpoint (v2)



2.4.2 Setting Approved URL

The Approved URL is the DNS used to resolve the SSO requests. To set it up, do the following:

1. Sign in to the [Microsoft Entra admin center](#).
2. Browse to **Identity > Applications > App registrations > <your application> > Authentication**.
3. Under **Platform configurations**, select **Add a platform**.
4. In the pane that opens, select the appropriate platform for your application. For example, select **Web** for a web application.
5. Under **Redirect URIs**, add the redirect URI of your application: https://<Maestro_instance_DNS>/maestro/dashboard, where <Maestro_instance_DNS> is the DNS for the instance on which Maestro application is deployed.
6. In **Supported Accounts type**, specify the scope of accounts for which SSO should work. The recommended option is “Accounts in this organizational directory only (... - Single tenant)”:

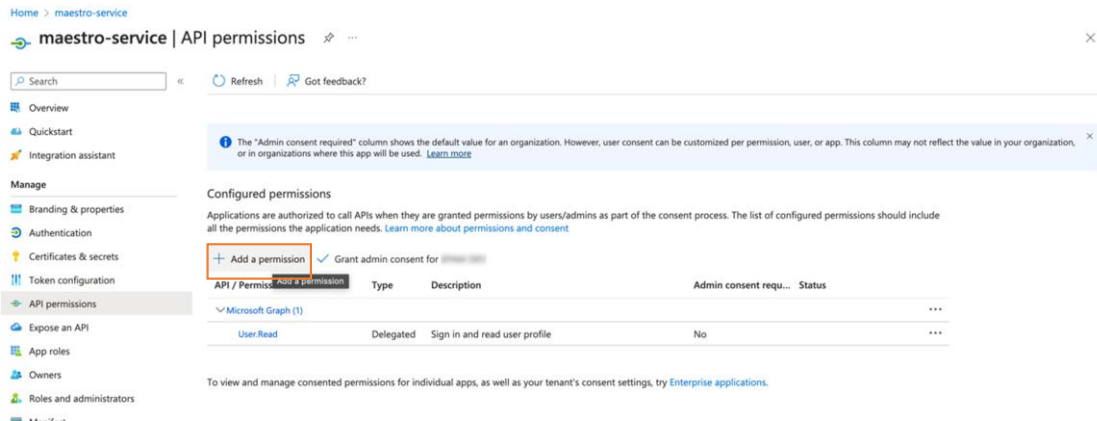


2.5 ADD API PERMISSIONS

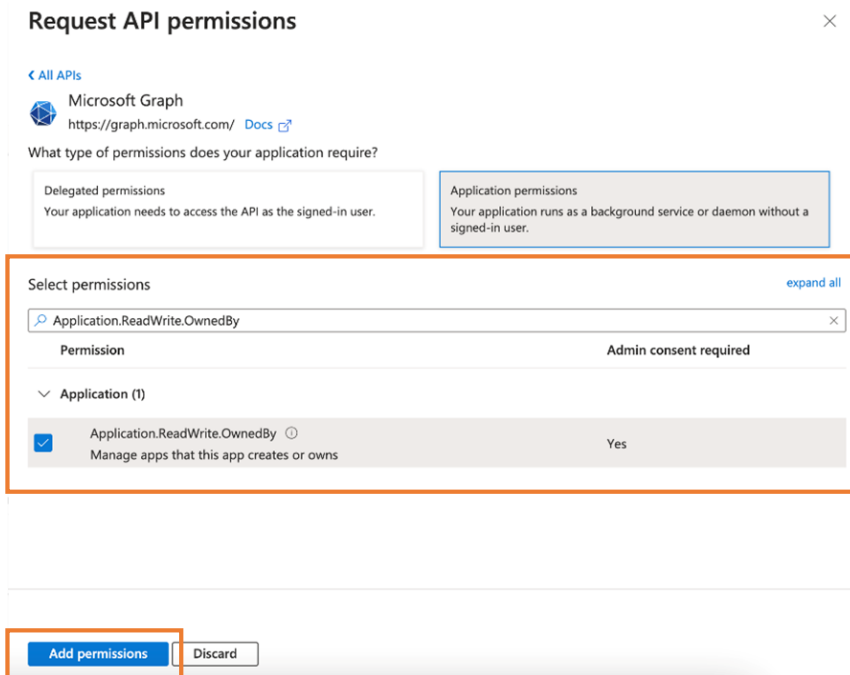
To enable API access to Maestro, do the following:

Maestro – Enabling Access to Azure Subscriptions

1. Open API permissions blade and select **Add Permission**:



2. Click **Request API permissions -> Microsoft Graph->Application Permissions->Application.ReadWrite.OwnedBy** and click **Add Permissions**.



3. In the API Permissions blade, click **Grant Admin Consent for <Organization>**:

Maestro – Enabling Access to Azure Subscriptions

Home > maestro-service

maestro-service | API permissions

Search Refresh Got feedback?

Overview

Quickstart

Integration assistant

Manage

Branding & properties

Authentication

Certificates & secrets

Token configuration

API permissions

Expose an API

App roles

Owners

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. The list of configured permissions should include all the permissions the application needs. [Learn more about permissions and consent](#)

+ Add a permission

Grant admin consent for [redacted]

API / Permissions name	Type	Grant admin consent for	Admin consent requ...	Status
Microsoft Graph (2)				
Application.ReadWrite.OwnedBy	Application	Manage apps that this app creates or owns	Yes	Not granted for [redacted]

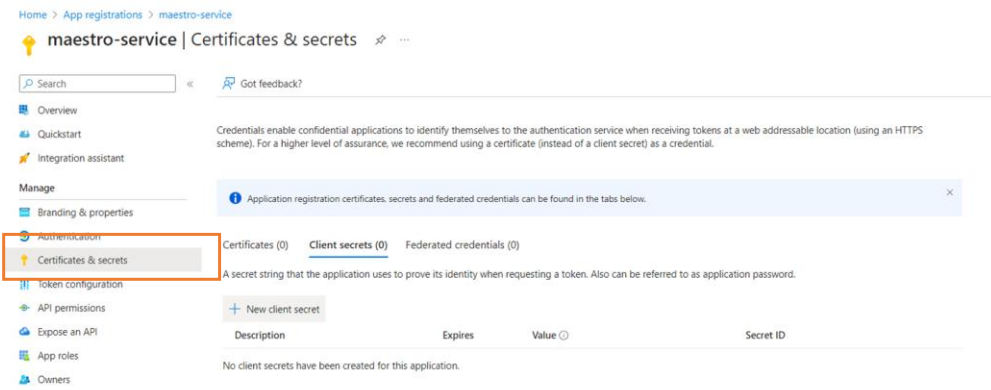
2.6 GENERATE A CLIENT SECRET FOR THE MICROSOFT ENTRA ID APPLICATION:

For Maestro to use specified Entra ID applications, two authorization options are possible: via client secrets or certificates.

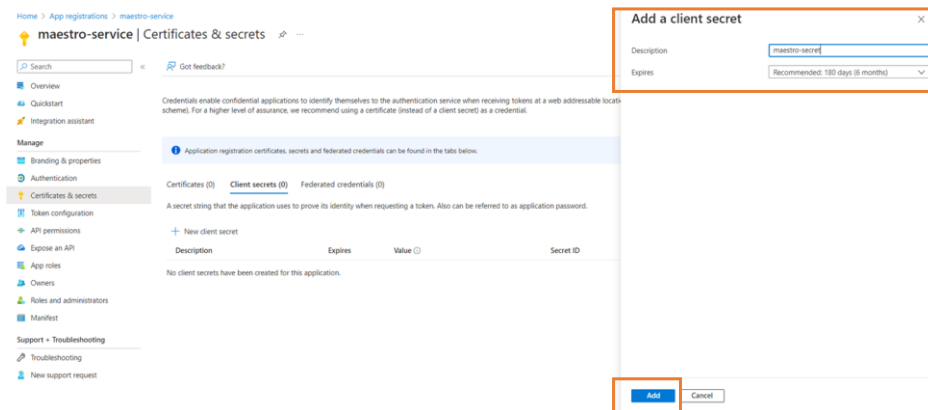
Using client secrets is more common.

To setup a client secret to authorize Service principal, do the following:

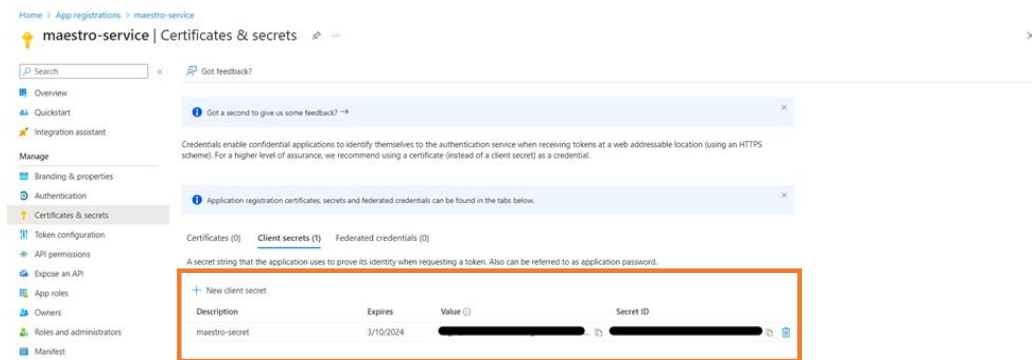
1. Open **Certificates & secrets** blade for the created application on **Azure Portal**



2. In the **Certificates and Client Secrets** tab click **New client secret**.
3. Enter the meaningful description and select the Expiry Date for the client secret to be created and click **Add**:



4. Review the generated secret:





The generated secret will be shown only once. Please copy the secret value and treat it as confidential.

3 SHARE THE CONNECTION DETAILS WITH MAESTRO TEAM

In a secure letter, share the following with the Maestro team:

- 1 Directory (tenant) ID (received at step 2.1).
- 2 Application (client) ID (received at step 2.1). If you use several applications, provide all IDs.
- 3 The ID of the storage account used for billing export (received at step 2.2.1)
- 4 The name of the container used for billing export (received at step 2.2.1)
- 5 The name of the role for the billing application (received at step 2.2.2)
- 6 Secret ID and Value (received at step 2.5) for each application you use.
- 7 Add all subscription IDs for which you want Maestro to have the access.
- 8 URL Endpoints (received at step 2.3.1)

VERSION HISTORY

Version	Date	Summary
3.3	February 23, 2026	Reviewed
3.2	October 28, 2024	Billing Export Setup added
3.1	February 19, 2024	Connection details reviews
3.0	August 22, 2023	Restructured
2.0	July 3, 2020	Combined from two initial docs
1.1	November 1, 2019	Chapter “Register Application in Azure AD” updated Page headers added
1.0	April 16, 2019	Combined from two initial docs